Kubning qirrasini topish

Kubning hamma qirralari bir xil uzunlikda. Shuning uchun kubning hajmi yoki bitta yog'ining yuzi ma'lum bo'lsa, qirrasini topa olamiz.

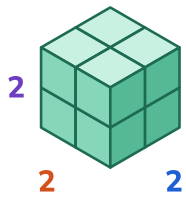
1-QISM

Mavzu

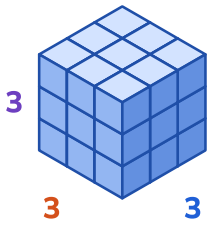
ESLAB OLAYLIK

- Kubning 6 ta yog'i — bir xil kvadratlar, 12 ta qirrasi esa teng.
- Kvadratning yuzi = tomoni × tomoni. Masalan, $4 \cdot 4 = 16$ cm².
- Kuboid hajmi = bo'yi × eni × balandligi. Kubda bu uchala o'lcham bir xil.

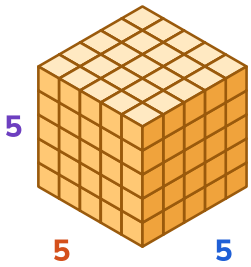
Rasmdagi uchta kub bir xil kubchalardan terilgan. A kubning qirrasi bo'ylab 2 ta kubcha turibdi, B kubnikida 3 ta, C kubnikida 5 ta. Har bir kub nechta kubchadan iborat?



A



B



C

Kub	Qirrasidagi kubchalar	Hajmi (kubchalar soni)
A	2	$2 \cdot 2 \cdot 2 = 8$
B	3	$3 \cdot 3 \cdot 3 = 27$
C	5	$5 \cdot 5 \cdot 5 = 125$

QOIDA · KUBNING HAJMI

Kub qirralarining uzunligi bir xil bo'lgani uchun:

Qirrasi × qirrasi × qirrasi = kubning hajmi



Kubning bitta yog'i yuzi esa qirrasi × qirrasi. Qirrasi 1 dan 12 gacha bo'lgan kublarni jadvalga yozib chiqdim.

Qirrasi	1	2	3	4	5	6
Bir yog'ining yuzi	1	4	9	16	25	36
Hajmi	1	8	27	64	125	216

Qirrasi	7	8	9	10	11	12
Bir yog'ining yuzi	49	64	81	100	121	144
Hajmi	343	512	729	1000	1331	1728

Jadvalni pastdan yuqoriga o'qisak, yog'ining yuzi yoki hajmidan qirra topiladi. Masalan, hajmi 729 bo'lsa, qirrasi 9 ga teng.

QOIDA · KVADRAT ILDIZ

Kvadratning tomoni yuzining kvadrat ildiziga teng.

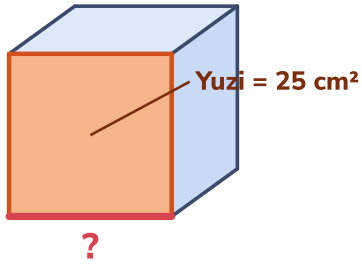
$$\textbf{Kvadratning tomoni} = \sqrt{\text{yuzi}}$$

$\sqrt{\ }$ — kvadrat ildiz. $\sqrt{324}$ — «kvadrat ildiz ostida 324» deb o'qiladi.

Kubning har bir yog'i — kvadrat, uning tomoni esa kubning qirrasi. Shuning uchun yog'ining yuzidan kvadrat ildiz olsak, kubning qirrasi chiqadi.

1-MISOL · JADVALDAGI SON

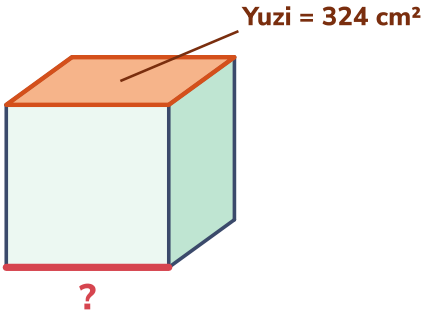
Kubning bir yog'i bo'yaldi, bo'yalgan yuza 25 cm². Qirrasi necha santimetr?



1. Qaysi son o'ziga ko'paytirilsa, 25 chiqadi? Jadvaldan: $5 \cdot 5 = 25$.
2. $\sqrt{25} = 5$, demak qirrasi 5 cm.

2-MISOL · JADVALDA YO'Q SON

Kubning ustki yog'i bo'yaldi, bo'yalgan yuza 324 cm². Qirrasi necha santimetr?



1. $10 \cdot 10 = 100$, $20 \cdot 20 = 400$. 324 ular orasida, demak qirrasi 10 dan uzun, 20 dan qisqa.
2. 324 soni 400 ga yaqin, shuning uchun 20 dan biroz kichik sonlarni sinaymiz: $19 \cdot 19 = 361$ — ko'p; $18 \cdot 18 = 324$ — to'g'ri keldi.
3. $\sqrt{324} = 18$, demak qirrasi 18 cm. Kalkulyatorda: 324 ni yozib, $\sqrt{}$ tugmasini bosamiz.

QOIDA · KUB ILDIZ

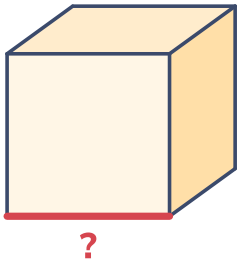
Kubning qirrasi uzunligi kub ildiz ostidagi hajmga teng.

$$\textbf{Kubning qirrasi} = \sqrt[3]{\text{hajm}}$$

$\sqrt[3]{}$ — kub ildiz belgisi. $\sqrt[3]{2197}$ — «kub ildiz ostida 2197» deb o'qiladi.

3-MISOL · JADVALDAGI SON

Kub shaklidagi yog'och bo'lakning hajmi 1000 cm³. Uning qirrasi necha santimetr?



1. Qaysi sonni son × son × son qilsak, 1000 chiqadi? Jadvaldan: $10 \cdot 10 \cdot 10 = 1000$.
2. $\sqrt[3]{1000} = 10$, demak qirrasi 10 cm.

4-MISOL · JADVALDA YO'Q SON

Kub shaklidagi don omborining hajmi 2197 m³. Omborning balandligi qancha?

1. Ombor kub shaklida, demak balandligi — kubning qirrasi. Jadvaldagi eng katta hajm — 1728, uning qirrasi 12. Bizning hajm kattaroq, demak qirrasi 12 dan uzun.
2. Sinaymiz: $13 \cdot 13 = 169$, $169 \cdot 13 = 2197$ — to'g'ri keldi.
3. $\sqrt[3]{2197} = 13$, demak balandligi 13 m. Kalkulyatorda: 2197 ni yozib, $\sqrt[3]{\ }$ tugmasini bosamiz.
4. Tekshirish: $13 \cdot 13 \cdot 13 = 2197$ m³.

USUL · KUBNING QIRRASINI TOPISH

1. Yog'ining yuzi berilsa: qirrasi × qirrasi = yuzi, demak qirrasi $= \sqrt{\text{yuzi}}$.
2. Hajmi berilsa: qirrasi × qirrasi × qirrasi = hajmi, demak qirrasi $= \sqrt[3]{\text{hajm}}$.
3. Kalkulyatorda: sonni yozib, $\sqrt{\ }$ yoki $\sqrt[3]{\ }$ tugmasini bosamiz.
4. Tekshirish: yuz uchun qirra × qirra, hajm uchun qirra × qirra × qirra hisoblanadi.

Kalkulyator yo'q bo'lsa, 10, 20, 30 ni sinab ko'ring: javob qaysi ikki son orasida ekani bilinadi, keyin oradagi sonlarni sinaysiz.

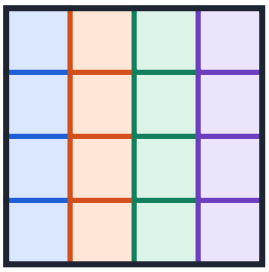


2-QISM

Misollar ishlash

1-MISOL · KVADRAT BO'LAKCHALARGA BO'LINGAN

Kvadrat shaklidagi shokolad 16 ta bir xil kvadrat bo'lakchadan iborat. Shokoladning yuzi 256 cm². Bitta bo'lakchaning tomoni qancha?



Yuzi = 256 cm²

1. **1-usul.** Shokoladning tomoni: $\sqrt{256} = 16$ cm, chunki $16 \cdot 16 = 256$. Har tomonda 4 ta bo'lakcha bor: $16 : 4 = 4$ cm.
2. **2-usul.** Bitta bo'lakchaning yuzi: $256 : 16 = 16$ cm². Uning tomoni: $\sqrt{16} = 4$ cm.

Ikkala usulda ham javob bir xil: 4 cm.

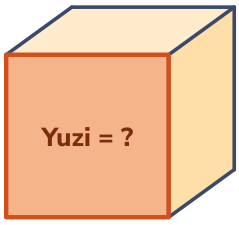
2-MISOL · KALKULYATORSIZ KUB ILDIZ

$\sqrt[3]{4913}$ ni toping.

1. O'nliklarni sinaymiz: $10 \cdot 10 \cdot 10 = 1000$, $20 \cdot 20 \cdot 20 = 8000$. 4913 ular orasida, demak javob 10 bilan 20 orasida.
2. O'rtadagi sondan boshlaymiz: $15 \cdot 15 = 225$, $225 \cdot 15 = 3375$ — kam. Javob 15 dan katta.
3. $17 \cdot 17 = 289$, $289 \cdot 17 = 4913$ — to'g'ri keldi.
4. $\sqrt[3]{4913} = 17$.

3-MISOL · HAJMDAN YOG'INING YUZINI TOPISH

Kub shaklidagi qutining hajmi 512 cm³. Qutining bir yog'iga yorliq yopishtirildi, u shu yog'ni to'liq qoplaydi. Yorliqning yuzi qancha?



1. Avval qirrasini topamiz: $8 \cdot 8 \cdot 8 = 512$, demak $\sqrt[3]{512} = 8$ cm.
2. Yorliq — tomoni 8 cm bo'lgan kvadrat: $8 \cdot 8 = 64$ cm².

Teskari savol bo'lsa, ya'ni yorliq yuzi 64 cm² berilib, hajm so'ralsa: $\sqrt{64} = 8$ cm, keyin $8 \cdot 8 \cdot 8 = 512$ cm³.



Qirrani topgach, uni o'ziga ko'paytirib ko'ring.
Berilgan yuz yoki hajm chiqmasa, javob noto'g'ri.

DIQQAT!

1. Hajmdan qirra topishda hajm 3 ga bo'linmaydi: ✗ $27 : 3 = 9$ ✓ $\sqrt[3]{27} = 3$, **chunki $3 \cdot 3 \cdot 3 = 27$**
2. Yog'ining yuzidan qirra topishda yuz 2 ga ham, 4 ga ham bo'linmaydi: ✗ $100 : 2 = 50$ ✗ $100 : 4 = 25$ ✓ $\sqrt{100} = 10$
3. Yuz berilsa — kvadrat ildiz, hajm berilsa — kub ildiz: ✗ yuzi 64 cm², qirrasi $\sqrt[3]{64} = 4$ cm ✓ **qirrasi $\sqrt{64} = 8$ cm**
4. Qirra — cm da, yuz — cm² da, hajm — cm³ da yoziladi: ✗ qirrasi 8 cm² ✓ **qirrasi 8 cm**

3-QISM

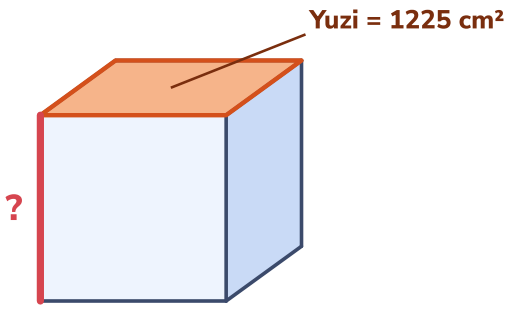
Mustahkamlash

1. Ko'paytmaga qarab ildizni yozing.
   a) $8 \cdot 8 = 64$, $\sqrt{64}$ b) $30 \cdot 30 = 900$, $\sqrt{900}$ c) $2 \cdot 2 \cdot 2 = 8$, $\sqrt[3]{8}$ d) $20 \cdot 20 \cdot 20 = 8000$, $\sqrt[3]{8000}$

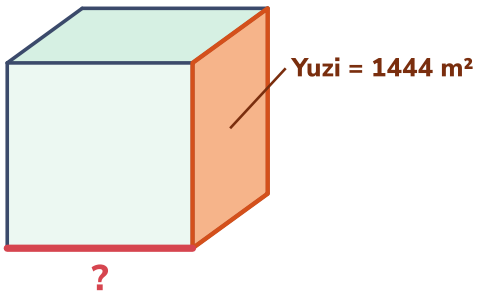
2. Kvadrat ildizni toping.
   a) $\sqrt{121}$ b) $\sqrt{441}$ c) $\sqrt{576}$ d) $\sqrt{1600}$

3. Kub ildizni toping. Kalkulyatordan foydalansa bo'ladi.
   a) $\sqrt[3]{6859}$ b) $\sqrt[3]{1\,000\,000}$ c) $\sqrt[3]{10\,648}$ d) $\sqrt[3]{24\,389}$

4. Rasmdagi kublarning qirrasi necha? Javobingizni qirra × qirra bilan tekshiring.

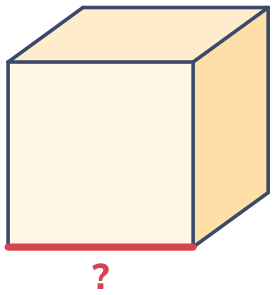


a)

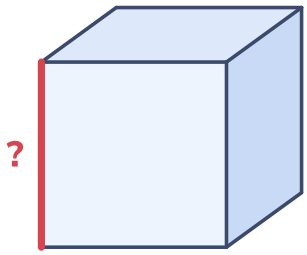


b)

5. Rasmdagi kublarning qirrasi necha? Javobingizni qirra × qirra × qirra bilan (yoki kalkulyatorda) tekshiring.

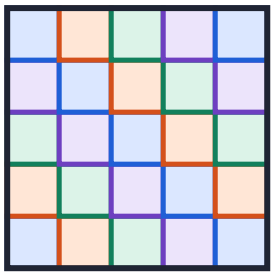


a) Hajmi 27 000 cm³



b) Hajmi 21 952 m³

6. Kvadrat shaklidagi devor panno 25 ta bir xil kvadrat plitkadan terilgan. Panno yuzi 2500 cm². Bitta plitkaning tomoni qancha?



Yuzi = 2500 cm²

7. Kub shaklidagi akvariumga 125 l suv sig'adi. Akvariumning balandligi qancha? ($1\ \text{l} = 1000\ \text{cm}^3$)

8. Kub shaklidagi sovg'a qutisining bir yog'i yuzi 676 cm².
   a) Qutining qirrasi qancha?
   b) Qutining hajmi qancha?

9. Ikkita kub bor. Birinchisining bir yog'i yuzi 784 cm², ikkinchisining hajmi 19 683 cm³. Qaysi kubning qirrasi uzunroq va necha cm ga?

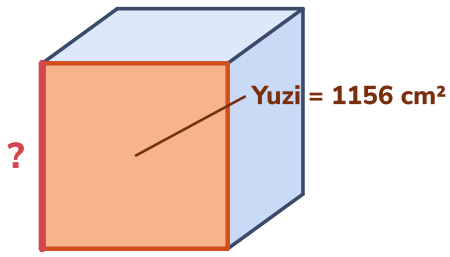
4-QISM

Mustaqil ish

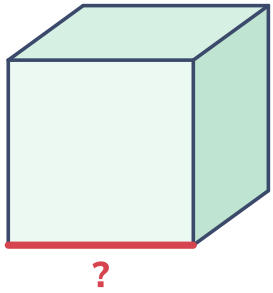
1. Kvadrat ildizni toping.
   a) $\sqrt{16}$ b) $\sqrt{841}$ c) $\sqrt{1296}$

2. Kub ildizni toping.
   a) $\sqrt[3]{27}$ b) $\sqrt[3]{12\,167}$ c) $\sqrt[3]{343\,000}$

3. Kubning qirrasini toping.



a)



b) Hajmi 32 768 m³

4. Kvadrat shaklidagi bog' 9 ta bir xil kvadrat gulzorga bo'lingan. Bog'ning yuzi 8100 m². Bitta gulzorning tomoni qancha?

5. Kub shaklidagi yashikning hajmi 216 000 cm³. Yashik qopqog'ining (ustki yog'ining) yuzi qancha?

6. Qirrasi 1 cm bo'lgan 1000 ta kubchadan bitta katta kub yasaldi.
   a) Katta kubning qirrasi necha cm?
   b) Katta kubning bitta yog'ida nechta kubcha ko'rinadi?

7. **Xatoni toping.** Kubning bir yog'i yuzi 484 cm². Shohruh qirrani shunday topdi: $484 : 4 = 121$ cm. Xato nimada? To'g'ri javob qancha?

5-QISM

Javoblar

Mustahkamlash

1-masala

a) $\sqrt{64} = 8$. b) $\sqrt{900} = 30$. c) $\sqrt[3]{8} = 2$. d) $\sqrt[3]{8000} = 20$.

2-masala

a) 11, chunki $11 \cdot 11 = 121$. b) 21, chunki $21 \cdot 21 = 441$. c) 24, chunki $24 \cdot 24 = 576$. d) 40, chunki $40 \cdot 40 = 1600$.

3-masala

a) 19, chunki $19 \cdot 19 \cdot 19 = 6859$. b) 100, chunki $100 \cdot 100 \cdot 100 = 1\,000\,000$. c) 22, chunki $22 \cdot 22 \cdot 22 = 10\,648$. d) 29, chunki $29 \cdot 29 \cdot 29 = 24\,389$.

4-masala

a) $\sqrt{1225} = 35$ cm; tekshirish: $35 \cdot 35 = 1225$. b) $\sqrt{1444} = 38$ m; tekshirish: $38 \cdot 38 = 1444$.

5-masala

a) $\sqrt[3]{27\,000} = 30$ cm; tekshirish: $30 \cdot 30 \cdot 30 = 27\,000$. b) $\sqrt[3]{21\,952} = 28$ m; tekshirish: $28 \cdot 28 \cdot 28 = 21\,952$.

6-masala

Panno tomoni: $\sqrt{2500} = 50$ cm. Har tomonda 5 ta plitka: $50 : 5 = 10$ cm. (Yoki: plitka yuzi $2500 : 25 = 100$ cm², $\sqrt{100} = 10$ cm.)

7-masala

$125\ \text{l} = 125\,000$ cm³. $\sqrt[3]{125\,000} = 50$ cm, chunki $50 \cdot 50 \cdot 50 = 125\,000$.

8-masala

a) $\sqrt{676} = 26$ cm. b) $26 \cdot 26 \cdot 26 = 17\,576$ cm³.

9-masala

Birinchisi: $\sqrt{784} = 28$ cm. Ikkinchisi: $\sqrt[3]{19\,683} = 27$ cm. Birinchi kubning qirrasi $28 - 27 = 1$ cm ga uzunroq.

Mustaqil ish

1-masala

a) 4. b) 29, chunki $29 \cdot 29 = 841$. c) 36, chunki $36 \cdot 36 = 1296$.

2-masala

a) 3. b) 23, chunki $23 \cdot 23 \cdot 23 = 12\,167$. c) 70, chunki $70 \cdot 70 \cdot 70 = 343\,000$.

3-masala

a) $\sqrt{1156} = 34$ cm. b) $\sqrt[3]{32\,768} = 32$ m.

4-masala

Bog'ning tomoni: $\sqrt{8100} = 90$ m. Har tomonda 3 ta gulzor: $90 : 3 = 30$ m.

5-masala

Qirrasi: $\sqrt[3]{216\,000} = 60$ cm. Qopqog'i yuzi: $60 \cdot 60 = 3600$ cm².

6-masala

a) $\sqrt[3]{1000} = 10$ cm. b) $10 \cdot 10 = 100$ ta.

7-masala

Shohruh yuzni 4 ga bo'lgan. Kvadrat ildiz olish kerak: $\sqrt{484} = 22$ cm, chunki $22 \cdot 22 = 484$.